SHUBHAM JOSHI

(202) 440-8445 | Email | Portfolio | GitHub | LinkedIn | Hugging Face

DATA SCIENTIST / APPLIED ML ENGINEER

Data scientist who turns unreliable data and ambiguous labels into defensible decisions and deployed models. Led the analytical core of a national healthcare engagement, caught a 12x data error, and improved a fine-tuned routing model from 52.3% to 73.1% through error analysis and task redesign.

SELECTED WORK

OpsPilot

Python, PyTorch, ModernBERT, Transformers, FastAPI, PostgreSQL · Model · GitHub

- Fine-tuned and published a ModernBERT support-ticket router on 23,747 leakage-safe examples; raised validation accuracy from 52.3% to 73.1% (weighted-F1 0.713) by redesigning ten overlapping labels into seven operational queues.
- Built confidence-gated routing that auto-routes 62.3% of tickets at 85.5% accuracy and returns uncertain cases for human review; rejected numerically unstable DeBERTa runs instead of reporting invalid gains.

RxCheck

Python, FastAPI, PostgreSQL, React, deterministic retrieval, structured model output · GitHub

- Imported and normalized 152,416 drug-interaction records with exact, brand, fuzzy, and NDC matching; kept detection deterministic while restricting the model to grounded explanations. Passed 26/26 formative scenarios.

AgentDock

TypeScript, Next.js, Prisma, PostgreSQL, Anthropic/OpenAI, MCP · GitHub · Demo

- Built a governed agent control plane that compiles natural-language goals into validated multi-step tool workflows with scoped permissions, approval gates, cost/step limits, and an append-only audit trail.
- Engineered durable execution with a PostgreSQL job queue, worker leases, crash recovery, idempotent external actions, step-cursor resume, SSE events, encrypted BYO keys, and approval-gated Gmail sends.

CLIENT DELIVERY

ACHP

Analytics Consultant · Johns Hopkins capstone · Alliance of Community Health Plans · 2025-2026

- Led most analytical work on a six-person client engagement; built a reproducible pipeline joining changing CMS schemas into a documented 138,263-row, 208-column master dataset and caught a 12x enrollment overcount before modeling.
- Used enrollment-weighted comparisons, OLS with HC3 robust errors, validated segmentation, county scoring, and plan-exit analysis; showed ACHP held the stronger cost position in 493 of 875 scored counties.
- Presented directly to ACHP leadership on a top-three schoolwide team; the client said the work would change where the organization focused its time.

LEADERSHIP & ADDITIONAL EXPERIENCE

Co-Founder & Vice President · Business Analytics & AI Organization, Johns Hopkins · 2025-2026

- Co-led a 100+ member organization; designed and delivered a three-hour applied-AI workshop and translated agents, RAG, evaluation, safety, and human oversight for mixed technical audiences.

Founder · Digital Marketing & Web Studio · Japan / Remote · 2021-2025

- Shipped websites and campaigns for international clients, owning delivery, client communication, audience research, A/B testing, and performance reporting across time zones.

TECHNICAL SKILLS

Modeling: Python, pandas, NumPy, scikit-learn, PyTorch, TensorFlow, Transformers, XGBoost, LightGBM, Optuna, regression, classification, clustering, calibration, SHAP/LIME

Statistics & evaluation: OLS/logistic regression, HC3 robust errors, hypothesis testing, cross-validation, ROC-AUC, macro/weighted F1, error analysis, leakage prevention, confidence thresholds

Data & delivery: PostgreSQL, SQL, ETL/ELT, multi-source integration, data quality, FastAPI, Tableau, Git, testing, CI/CD, Hugging Face, stakeholder communication

EDUCATION & CREDENTIALS

Johns Hopkins University, Carey Business School · M.S., Business Analytics & Artificial Intelligence · Completed July 2026 · Dean's Scholarship

Nagoya University of Commerce & Business · B.B.A., Global Business Administration · 2025 · Dean's Scholar · Exchange: Monash University

Certifications: Microsoft Azure AI Fundamentals (AI-900) · SAS SQL Essentials · SAS AI Foundations **Languages:** English, Japanese, Nepali (native) · Hindi (fluent)